Entropy Rectifying Guidance for Diffusion and Flow Models

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Introduction: The Guidance Problem in Diffusion Models

What is guidance?

- Diffusion models generate images by iterative denoising from noise to image
- **Guidance** = steering the generation toward desired outputs (e.g., matching a text prompt)
- Standard method: **Classifier-Free Guidance (CFG)**

The fundamental problem:

Improving one aspect hurts the others

Quality	Diversity	Consistency
Are images realistic?	Different outputs per prompt?	Match the prompt?

CFG: Better quality & consistency $\rightarrow$ worse diversity + over-saturation

Solution: Entropy Rectifying Guidance (ERG)

Main Idea

Create a "weaker" version of the model by manipulating its attention layers at inference time, no second model, no retraining.

Standard CFG:

Strong − Unconditional

- Needs unconditional training
- Quality-diversity trade-off
- Doesn't work for unconditional sampling

ERG:

Strong − Weakened attention

- ✓ No extra training needed
- ✓ Improves quality, diversity **and** consistency
- ✓ Works for any mode
- ✓ Same compute as CFG

How It Works: Manipulating the Attention Energy

Key insight: Attention = one gradient step minimizing a Hopfield energy

Standard Energy

$$E(\xi; X) = \frac{1}{2}\xi^\top \xi - \mathrm{LSE}(X^\top \xi,\ \beta)$$

ERG Modified Energy

$$E(\xi; X) = \frac{1}{2}\xi^\top \xi - \alpha \cdot \mathrm{LSE}(X^\top \xi,\ \tau \cdot \beta)$$

4 hyperparameters:

- τ: attention temperature (sharpness)
- α: pattern matching weight
- λ: gradient step size
- K: number of update iterations

Default: $\alpha = \lambda = \tau = K = 1$ recovers standard attention.

Algorithm: ERG Attention Update

Input: Q, K, V, parameters α, τ, λ, K

for $k = 1$ to K **do**:

$$Q \leftarrow Q - \lambda\Big(Q - \alpha \cdot \mathrm{softmax}\big(\tau \cdot QK^\top\big) V\Big)$$

end for

Return: Q (replaces standard attention output)

In practice, $K = 1, \lambda = 1$ works well $\rightarrow$ reduces to:

$$Q' = \alpha \cdot \mathrm{softmax}(\tau \cdot QK^\top)\, V$$

ERG: Putting It All Together

Apply energy rectification at two places:

1. **C-ERG (Text Encoder):** temperature τ_c in self-attention $\rightarrow$ weakened embeddings c_τ
2. **I-ERG (Image Denoiser):** parameters $\{\alpha, \lambda, \tau, K\}$ in attention layers $\rightarrow$ weaker denoiser D_ξ
   - Applied only after kickoff threshold ϕ (preserve early global structure)
   - Applied to selected layers, not all

Final ERG Update Rule

$$\Psi_{\text{ERG}}(x, c, t) = w \cdot \underbrace{D(x,\ c,\ t)}_{\text{normal model}} + (1-w) \cdot \underbrace{D_\xi(x,\ c_\tau,\ t;\ \Theta_\xi)}_{\substack{\text{weakened attention} \\ +\ \text{weakened text}}}$$

Key advantages:

- Replaces the unconditional term in CFG $\rightarrow$ works even without unconditional training
- Can be combined with CADS and APG for further improvements
- No retraining, works on any pre-trained attention-based diffusion/flow model

Experiments: Setup

- **Tasks:**
  - Text-to-image
  - Unconditional generation
  - Class-conditional generation
- **Models:**
  - DiT XL/2 (class-conditional)
  - MMDiT-like model (text-to-image)
  - Llama3-8B + Flan-T5-XL text encoders
- **Metrics:**
  - Quality: FID, Density
  - Diversity: Coverage
  - Consistency: CLIPScore, VQAScore
- **Protocol:** 50 Euler steps, compared with CFG and recent guidance baselines

Text-to-Image Results (COCO'14)

Method	FID↓	Density↑	Coverage↑	CLIP↑	VQA↑	NFE
CFG	12.81	98.24	71.12	26.45	70.15	2
APG	11.88	104.07	73.06	26.54	72.47	2
CADS	11.93	101.01	72.99	26.76	73.36	2
PAG*	12.75	107.21	72.20	26.80	73.32	2
SAG*	11.68	103.58	72.74	26.81	72.16	2
SEG*	16.87	87.77	61.91	26.86	73.59	3
AutoGuidance	16.62	87.02	62.75	26.59	73.53	2
ERG	13.62	**120.25**	73.21	**26.86**	**73.96**	2
ERG + APG	**11.37**	115.08	**80.50**	26.74	73.55	2
ERG + CADS	12.87	**128.54**	76.23	26.75	73.45	2

- ERG significantly improves **sample quality (Density)** and **prompt consistency**.
- Combining ERG with APG gives the **best diversity (Coverage)**.
- ERG requires only **2 model evaluations per step** (same cost as CFG).

Text-to-Image: Trade-offs (Pareto Fronts)

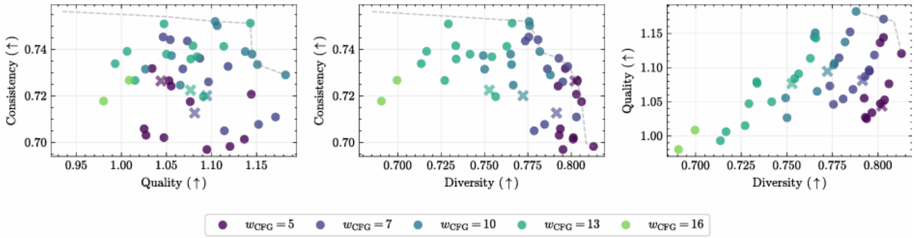

Figure 2: Pareto fronts on Quality (Density), Diversity (Coverage), Consistency (VQAScore).

Experiment idea

- Vary the guidance strength w
- Vary ERG parameters (α, τ)
- Compare **APG** with **ERG + APG**

Key observation

- **ERG + APG achieves better trade-offs**
- Higher image quality, diversity, and prompt consistency

Unconditional Sampling: Qualitative + Quantitative

Figure 3: No guidance vs ERG (empty prompt).

Table 2 (Unconditional)

Method	FID↓	Density↑	Coverage↑
No guidance	101.50	8.99	3.63
SEG*	37.75	55.56	34.79
AutoGuidance	39.50	48.26	34.71
ERG	**36.25**	**55.84**	**51.59**

Observations

- ERG works even **without conditioning**
- Large improvement in **diversity** (Coverage)
- Better global structure and realism

Class-Conditional + Ablations

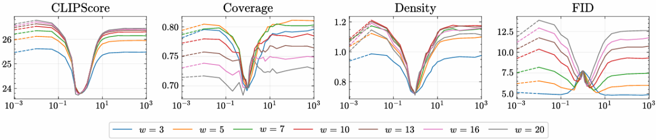

Figure 4: Effect of τ_c on consistency/diversity trade-off.

Class-conditional (ImageNet)

- ERG improves **Density** at 256 and 512
- At 512, ERG also gives the **best FID**
- Coverage remains close to CFG

Ablations

- τ_c: mainly improves **consistency**
- τ_i: mainly improves **quality/diversity**
- Multiple inner steps give little gain $\Rightarrow K = 1$

Conclusion

- **Entropy Rectifying Guidance (ERG)** improves the trade-off between image **quality**, **diversity**, and **prompt consistency**.
- ERG creates a **weak guidance signal** by modifying the attention mechanism, without requiring an additional model or retraining.
- The method works for **text-to-image, class-conditional, and unconditional generation** while keeping the **same computational cost as CFG**.